

How do you like your dairy?

A review of how the world tastes milk products

s1021142

Radboud University, Nijmegen

ABSTRACT

1 INTRODUCTION

Dairy consumption varies across cultures. For example, ethnicity affects the liking of yogurt [9]. Moreover, the same dairy products can be perceived as differing in healthiness across cultures [4]. In addition, country characteristics can influence which dairy products are consumed [8]. Therefore, studying the relationship between people and dairy can clarify the broader human relationship with food and deepen the understanding of cultural differences.

Research question. What is the relationship between dairy products and the country of origin?

- For each dairy product, where do they originate from?
- How is the dairy product consumed?
- Is there a relationship between the dairy product and the country of origin?

2 RELATED WORK

Dairy perception and consumption are strongly shaped by culture. At the same time, the history and geography of dairy farming influence how dairy products are produced and valued, and the country of origin of a product can influence consumers. This section goes into more depth on these topics and further discusses food computing and the current research.

2.1 Dairy and culture

A number of studies show that people in different countries perceive and like dairy products in different ways. For example, work on yogurt with different degrees of granularity [9] found that Chinese and Danish consumers responded differently to the presence or absence of particles. Moreover, Johansen et al. [4] studied young Norwegian, Danish, and Californian consumers and showed that all three groups have common motives for consuming calorie-reduced dairy products (e.g. fat content, perceived healthiness and taste), but their relative importance differs across countries. In addition, research on Australian and Chinese consumers [8] reports cross-country differences in the liking of pasteurized milk versus long shelf life milk and in how shelf life labeling affects liking. Lastly, research with a trained panel on flavor lexicons for cheese [1] showed that the panels from New Zealand, Ireland, and the United States used different words to describe the same cheeses, and tended to group cheeses by country of origin.

Beyond sensory perception, cultural and historical factors also influence how dairy is produced, valued, and consumed. For example, in India most milk comes from the buffalo rather than cows because

cows have a sacred status in Hinduism. Moreover, in the United States, people are increasingly shifting towards plant-based milks [17]. Lastly, dairy farming can be of great importance to mountain communities [15], family farming cultures [3], and animal welfare [14]. This is further emphasized by Valenze [16], who researched the different religious meanings of milk and the different functions throughout the ages. Thus, dairy is not only a food product, but also has cultural, religious, and socio-economic significance.

2.2 Country-of-origin labeling

Country-of-origin (COO) research has shown that the country where a product is made can influence how consumers evaluate it. This suggests that links between dairy products and countries can shape perceptions and choices. Knight and Calantone [5] propose a model of how COO information is processed cognitively and show that COO effects are complex and cultural context plays an important role in purchase decisions. Moreover, research on the Canadian market [12] found that COO labeling affects the purchase choice and willingness to pay for dairy products. Finally, Yang et al. [19] illustrate how consumer behavior is influenced through COO information after a major food-related scandal.

2.3 Food computing

Food computing and computational gastronomy are emerging research areas that use computational methods to study food-related topics. Surveys and overviews in this area [2, 11] describe how food data (e.g. recipes, images, social media posts) can be analyzed with tools from data science and artificial intelligence to address a wide range of questions about flavor, health, and culinary culture. Several studies focus on culinary cultures and food-country relations. For example, Laufer et al. [7] mine cultural relations between different language communities in Europe by analyzing how food cultures are described on Wikipedia pages across different language editions. Moreover, palette varieties around the world are investigated [13] and are found to have strong geographical and cultural similarities between countries. Furthermore, a large visual question answering dataset with text-image pairs is made identifying dish names and their origins [18]. Lastly, two topic modelling approaches are applied to recipes [6, 10] to discover more about the human relationship with food.

2.4 Current research

Previous research has shown that culture influences how dairy products are perceived and consumed, and that the country of origin of a product can affect how consumers evaluate it. Moreover, work on food computing has demonstrated that large-scale online data and computational methods can be used to study food-related topics, including culinary cultures and food-country relations. However, there is little work that combines these perspectives to study how

dairy products and countries are linked in textual descriptions. Therefore, this paper uses text mining on Wikipedia articles to investigate how dairy products are associated with their countries of origin.

3 APPROACH

3.1 Data

Wikipedia pages of dairy products and countries. All pages are linked in this Wikipedia page: https://en.wikipedia.org/wiki/List_of_dairy_products.

3.2 Method

The linked Wikipedia page provides a list of dairy product and origins, answering my first subquestion. In the data I can look for words (such as e.g. fermented, cooked, breakfast, bread) in paragraphs where a specific dairy product is mentioned to answer my second sub question. Lastly, I will search in the country pages for the dairy product and for the dairy product in the country pages, extracting those paragraphs and providing me with information about the relation between the country and the dairy product, answering my third sub question.

3.3 Validation

The output will be confirmed through human judgement and compared with previous research.

4 RESULTS

5 DISCUSSION

6 CONCLUSION

REFERENCES

- [1] M.A. Drake, M.D. Yates, P.D. Gerard, C.M. Delahunty, E.M. Sheehan, R.P. Turnbull, and T.M. Dodds. 2005. Comparison of differences between lexicons for descriptive analysis of Cheddar cheese flavour in Ireland, New Zealand, and the United States of America. *International dairy journal* 15, 5 (2005), 473–483.
- [2] Mansi Goel and Ganesh Bagler. 2022. Computational gastronomy: A data science approach to food. *Journal of biosciences* 47, 1 (2022), 12–12.
- [3] Ren Hu and Nicholas Gill. 2021. The Family Farming Culture of Dairy Farmers: A Case-Study of the Illawarra Region, New South Wales. *Sociologia ruralis* 61, 2 (2021), 398–421.
- [4] Susanne Bølling Johansen, Tormod Næs, and Margrethe Hersleth. 2011. Motivation for choice and healthiness perception of calorie-reduced dairy products. A cross-cultural study. *Appetite* 56, 1 (2011), 15–24.
- [5] Gary A. Knight and Roger J. Calantone. 2000. A flexible model of consumer country-of-origin perceptions: A cross-cultural investigation. *International marketing review* 17, 2 (2000), 127–145.
- [6] Maija Kåle and Ebenezer Agbozo. 2021. Utility of Large-Scale Recipe Data in Food Computing. *Baltic Journal of Modern Computing* 9, 2 (2021), 155–165.
- [7] Paul Laufer, Claudia Wagner, Fabian Flöck, and Markus Strohmaier. 2015. Mining cross-cultural relations from Wikipedia: A study of 31 European food cultures. In *Proceedings of the ACM Web Science Conference* (Oxford, United Kingdom) (*WebSci '15*). Association for Computing Machinery, New York, NY, USA, Article 3, 10 pages. doi:10.1145/2786451.2786452
- [8] D.G. Liem, D.P. Bolhuis, X. Hu, and R.S.J. Keast. 2016. Short communication: Influence of labeling on Australian and Chinese consumers' liking of milk with short (pasteurized) and long (UHT) shelf life. *Journal of dairy science* 99, 3 (2016), 1747–1754.
- [9] Jing Liu, Anne C. Bech, Sandra Stolzenbach Waehrens, and Wender L.P. Bredie. 2021. Perception and liking of yogurts with different degrees of granularity in relation to ethnicity, preferred oral processing and lingual tactile acuity. *Food quality and preference* 90 (2021), 104158–.
- [10] Weiqing Min, Bing-Kun Bao, Shuhuan Mei, Yaohui Zhu, Yong Rui, and Shuqiang Jiang. 2018. You Are What You Eat: Exploring Rich Recipe Information for Cross-Region Food Analysis. *IEEE transactions on multimedia* 20, 4 (2018), 950–964.
- [11] Weiqing Min, Shuqiang Jiang, Linhu Liu, Yong Rui, and Ramesh Jain. 2020. A Survey on Food Computing. *ACM computing surveys* 52, 5 (2020), 1–36.
- [12] Amanda Norris and John Cranfield. 2019. Consumer Preferences for Country-of-Origin Labeling in Protected Markets: Evidence from the Canadian Dairy Market. *Applied economic perspectives and policy* 41, 3 (2019), 391–403.
- [13] Sina Sajadmanesh, Sina Jafarzadeh, Seyed Ali Ossia, Hamid R. Rabiee, Hamed Haddadi, Yelena Mejova, Mirco Musolesi, Emiliano De Cristofaro, and Gianluca Stringhini. 2017. Kissing Cuisines: Exploring Worldwide Culinary Habits on the Web. In *Proceedings of the 26th International Conference on World Wide Web Companion*. International World Wide Web Conferences Steering Committee, Republic and Canton of Geneva, Switzerland, 1013–1021.
- [14] Michelle Sinclair, Natasha Y. P. Lee, Maria José Hötzel, Maria Catalina T. de Luna, Arvind Sharma, Musadiq Idris, Tessa Derkley, Congcong Li, Mohammad Ariful Islam, Oluwaseun S. Iyasere, Grisel Navarro, Abdelkareem A. Ahmed, Chanadda Khruapradab, Michael Curry, Georgette Leah Burns, and Jeremy N. Marchant. 2022. International perceptions of animals and the importance of their welfare. *Frontiers in animal science* 3 (2022).
- [15] Singh and Bohra. 2021. *Dairy Farming in Mountain Areas*. Daya Pub. House.
- [16] Deborah M. Valenze. 2011. *Milk : a local and global history* (1st ed. ed.). Yale University Press, New Haven, Connecticut.
- [17] Andrea S. Wiley. 2014. *Cultures of Milk : The Biology and Meaning of Dairy Products in the United States and India*. Harvard University Press, Cambridge, MA.
- [18] Genta Indra Winata, Frederikus Hudi, Patrick Amadeus Irawan, David Anugraha, Rifki Afina Putri, Wang Yutong, Adam Nohejl, Ubaidillah Ariq Prathama, Nedjma Ousidhoum, Afifa Amriani, Anar Rzaev, Anirban Das, Ashmari Pramodya, Aulia Adila, Bryan Willie, Candy Olivia Mawalim, Cheng Ching Lam, Daud Abolade, Emmanuele Chersoni, Enrico Santus, Fariz Ikhwantari, Garry Kuwanto, Hanyang Zhao, Haryo Akbarianto Wibowo, Holy Lovenia, Jan Christian Blaise Cruz, Jan Wira Gotama Putra, Junho Myung, Lucky Susanto, Maria Angelica Riera Machin, Marina Zhukova, Michael Anugraha, Muhammad Farid Adilazuarda, Natasha Christabelle Santosa, Peerat Limkonchotiwat, Raj Dabre, Rio Alexander Audino, Samuel Cahyawijaya, Shi-Xiong Zhang, Stephanie Yulia Salim, Yi Zhou, Yinxuan Gui, David Ifeoluwa Adelani, En-Shiun Annie Lee, Shogo Okada, Ayu Purwarianti, Alham Fikri Aji, Taro Watanabe, Derry Tanti Wijaya, Alice Oh, and Chong-Wah Ngo. 2025. WorldCuisines: A Massive-Scale Benchmark for Multilingual and Multicultural Visual Question Answering on Global Cuisines. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Luis Chiruzzo, Alan Ritter, and Lu Wang (Eds.). Association for Computational Linguistics, Albuquerque, New Mexico, 3242–3264. doi:10.18653/v1/2025.naacl-long.167
- [19] Rongbin Yang, Roshnee Ramsaran, and Santoso Wibowo. 2018. An investigation into the perceptions of Chinese consumers towards the country-of-origin of dairy products. *International journal of consumer studies* 42, 2 (2018), 205–216.

7 WORK REPORT

In order to find literature, I mainly looked in a online university library. I used queries such as 'milk', 'dairy', 'country of origin dairy', and 'food computing'.